Cross-Layer Alignment Network With Norm Constraints for Fault Diagnosis Under Varying Working Conditions

Xilin Yang, Xianfeng Yuan, Tianyi Ye, Qingyang Xu, Yong Song, and Jiong Jin, *Member, IEEE*

***Abstract*— The changes in rotating speed and load of mechanical systems cause the varying working conditions of gearboxes or bearings, so deep learning-based fault diagnosis models face inconsistent distribution of training and test sets, resulting in a great reduction in diagnosis accuracy. To solve this problem, various effective transfer learning-based fault diagnosis methods are proposed in the literature; however, the majority of current works still suffer the following drawbacks: First, they usually align the distribution at the same-level layer of the parameter-shared feature extractor, but in fact, the same-level fault features may distribute in different-level layers of the feature extraction network. Second, the model trained under supervised information of the source domain cannot guarantee discriminability for the target domain samples. To address the above issues, a novel unsupervised adaptive cross-layer alignment network with norm constraints (ACLAN-NC) framework is developed in this article, which aligns the same-level fault features among the same-level and different-level model layers. By using those cross-layer pairs, we can align the fault features at a fine-grained level and reduce inappropriate knowledge transfer. The batch nuclear-norm minimization and maximization technique is, moreover, used to restrict the prediction output of the source domain and target domain, ensuring the discriminability and diversity of the model. To demonstrate the effectiveness of ACLAN-NC for fault diagnosis under varying working conditions, extensive experiments are conducted to manifest the superior performance of our proposed ACLAN-NC framework.**

***Index Terms*— Batch nuclear norm, cross-layer alignment, intelligent fault diagnosis (IFD), unsupervised domain adaptation (UDA), varying working conditions.**

I. Introduction

MECHANICAL equipment plays an essential role in the manufacturing industry, and particularly, rotating components, such as gearboxes and rolling bearings, are the key components of various mechanical equipment; however, the harsh working environment of rotating components could cause serious performance degradation or failure of the equipment [1].

The signal processing-based fault diagnosis methods can extract fault-related characteristics and exhibit promising diagnosis performance [2], [3], while the end-to-end intelligent fault diagnosis (IFD) technique is a rapid-developing and ever-growing field, which can achieve excellent results without the requirement of manual features or advanced signal processing techniques [4], [5]. To extract fault features, an attention-based convolutional neural network is proposed in [6]. A structure composed of convolutional neural networks and a deep forest classifier is designed in [7] to fully exploit the features of time-frequency images. One of the basic assumptions of deep learning-based approaches, however, is that the test and training sets obey the independent identity distribution. The variations in rotating speed or load can, however, lead to varying working conditions in gearboxes or rolling bearings. Compared with constant working conditions, the vibration signals under varying conditions usually have the characteristics of nonconstant sampling phases, pulsating impact interval, amplitude variation, and signal noise pollution. The varying conditions lead to inconsistent distribution of the training and test sets, which can dramatically degrade the performance of IFD models. Adequate labeled data is another requirement for training IFD models, while collecting data in real industry scenarios is not always easy and manual labeling of fault data under changing operating conditions is time-consuming and expensive. The above drawbacks negatively affect the generalization performance and reduce the accuracy of IFD models.

Unsupervised domain adaptation (UDA) is able to perform well in an unlabeled target domain with the help of a labeled source domain [8], [9], which is introduced into transfer learning-based fault diagnosis. In [10], the marginal and conditional distribution discrepancy was minimized by a metric, i.e., maximum mean discrepancy (MMD). Similarly, Jiao et al. [11] reduced joint MMD (JMMD) distance between target and source domains to extract domain-invariant and discriminative fault features. Second-order statistical distance like correlation alignment (CORAL) [12] was combined with deep networks to generate the samples needed to train a

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fault diagnosis model [13]. In addition to the metric-based methods, the adversarial-based method is another mainstream approach to solving UDA problems. Ganin et al. [14] first proposed an architecture to extract domain-invariant features, where the domain discriminator and extractor were trained adversarially by a gradient reverse layer. Since then, there has been a proliferation of adversarial-based fault diagnosis architectures. A fault diagnosis approach was presented by Chen et al. [15], in which the source feature encoder and classifier were trained under label supervision, while the target feature encoder and domain discriminator were trained adversarially. Wasserstein distance [16] could better measure the distribution because it effectively avoids gradient disappearance and explosion problems in the adversarial training process. Wasserstein distance-based discriminator helps the extractor to learn features that were simultaneously discriminative and domain-invariant [17]. Different from the above approaches, Jiao et al. [18] exploited the class confusion minimizing method and restricted the prediction output of the target domain for good transfer performance. To reduce the domain divergence caused by different machines, [19] and [20] integrated deep transfer learning and the Bayesian neural networks to reduce distribution divergence. With the help of Monte Carlo dropout inference, Bayesian deep learning framework could quantify the uncertainty of prediction, and then transfer learning is embedded into the framework, achieving satisfactory cross-machine fault prognosis performance.

First, to learn domain-invariant fault features, most existing works rely on a fixed distribution metric or a domain discriminator to align features. The above works only aligned source and target domain feature distributions at the same-level layer of the parameter-shared network [21], as shown in Fig. 1(b). The underlying assumption is that the same-level fault features from different domains can be extracted in the same-level model layers. It should be noticed that the level of features refers to the deep degree. The layers close to the original input extract shallow and common features, while the deep and task-specific features are extracted in the layers near the end of the extractor. The level of the model layer means the total number of layers the feedforward features have passed by in this layer; however, according to [21] and [22], it is noted that feature information of the same level may be distributed between different layers of the model because of the domain divergence. Directly aligning the features of the same-level layers may lead to a feature dislocation problem because the same-level fault features are not in the same-level layers of the source extractor and target extractor, as shown in Fig. 1(b). There have been some works that focus on feature dislocation problems. For example, Chen et al. [22] proposed a semantic calibration for cross-layer knowledge distillation, which automatically assigns the appropriate target layers of the teacher model to each layer of the student model. Yuan et al. [23] proposed to use the cross-layer alignment to achieve collaborative feature aggregation and calibration and learn a fused model for domain generalization between separated domains.

Second, due to the distribution divergence, models trained under the supervision of source labels may lose

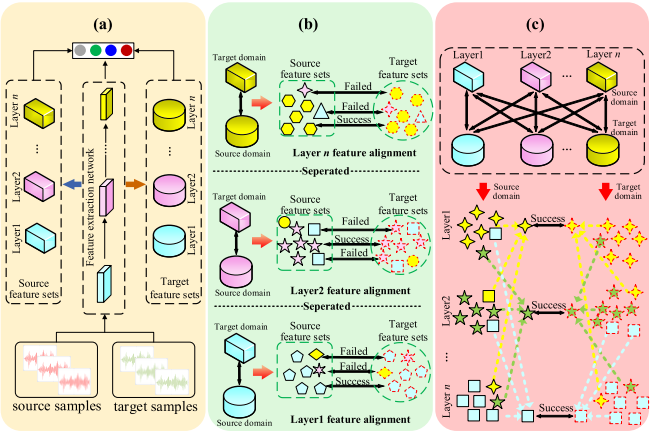

Fig. 1. Different domain alignment frameworks. The commonly used same-layer alignment methods directly match the same layer features, while cross-layer alignment matches the same-level features in different layers to further improve domain adaptation. (a) Multilevel feature. (b) Same-layer alignment. (c) Cross-layer alignment.

discriminability in the target domain, which causes the fault classifier to make uncertain predictions [24]. In addition, some fault types are dominant in a random batch, which makes the deep learning models tend to predict samples as the dominant category and degrades the diversity of fault diagnosis models [25]. To summarize, the lack of discriminability and diversity further reduces the fault diagnosis accuracy.

To overcome the mentioned problems and drawbacks, we design a cross-layer alignment network based on the residual structure, leveraging feature information at each level of the network for aligning source and target domains. Specifically, the source and target domain original vibration signals after segmentation are fed into the residual connected extractor, and multilevel fault features are obtained in different layers of the extractor. Then, the JMMD distance is used as the metric to measure whether the domain divergence between the same-level fault features is in the same-level layers or different-level layers. The batch nuclear-norm constraint technique is, moreover, introduced into the proposed framework to maintain the discriminability and diversity of the deep learning model.

To summarize, the main innovations and contributions are listed as follows.

1) An IFD framework that considers the fault feature dislocation problem is constructed. To address this issue, an adaptive domain alignment mechanism for cross-layer alignment is introduced into the proposed framework. Using this alignment mechanism, the same-level fault features of the source domain and target domain, whether distributed in the same-layer or cross-layer pair, will be automatically aligned.
2) For an excellent fault diagnosis performance under varying working conditions, a hybrid loss function is proposed, mainly including three parts: the cross-entropy classification loss, the alignment loss, and the batch nuclear-norm constraints loss. By using the designed loss function, we are able to strengthen the feature alignment and meanwhile yield a balance between classifier

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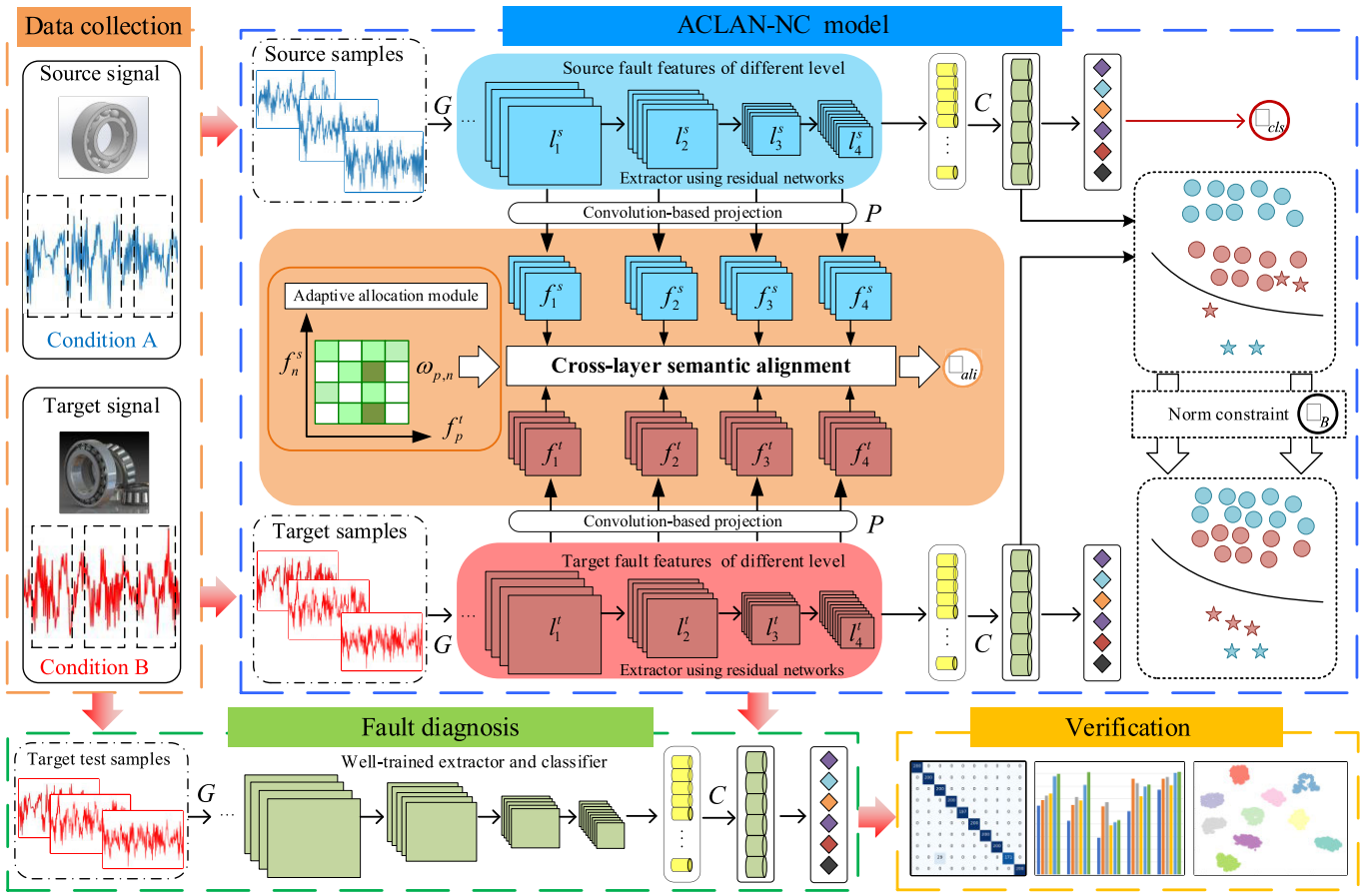

Fig. 2. Flowchart of the proposed fault diagnosis approach.

discriminability and diversity with the help of batch nuclear-norm constraints.

3) Various advanced domain adaptation methods are chosen to compare with adaptive cross-layer alignment networks with norm constraints (ACLAN-NC), and the effectiveness and superiority of ACLAN-NC are validated by the gearbox and bearing fault diagnosis experiments.

The remainder of this article is organized as follows. In Section II, the theoretical background will be introduced. Then, the proposed method is detailed in Section III. Section IV presents the results and analysis of the experiment. Finally, Section V concludes this article.

II. Preliminaries

A. Problem Definition

In this article, the source domain has n_s labeled samples while the target domain has n_t unlabeled target samples, denoted as $\mathcal{D}_s = \{(\mathbf{x}_1^s, y_1^s), \ldots, (\mathbf{x}_{n_s}^s, y_{n_s}^s)\}$ and $\mathcal{D}_t = \{\mathbf{x}_1^t, \ldots, \mathbf{x}_{n_t}^t\}$, respectively. $\mathbf{x}_i \in \mathcal{X}$ is the sample and $y_i \in \mathcal{Y}$ is the corresponding label. The healthy or faulty machine categories are the same for source and target domains, meaning that $\mathcal{Y}_s = \mathcal{Y}_t = \mathcal{C}$. IFD under UDA settings aims at designing and training a model that extracts discriminative and domain-invariant features so that the classifier could accurately identify samples.

B. Structure of Residual Network

Deep neural networks are difficult to converge, and the residual connection is qualified to address the problem of vanishing/exploding gradients that will damage the convergence of networks [26]. Residual learning aimed to make layers fit a residual mapping instead of a directly desired underlying mapping. Suppose $\boldsymbol{x}$ is the input, and the mapping of stacked nonlinear layers can be written as $\mathcal{F}(\boldsymbol{x}) := \mathcal{H}(\boldsymbol{x}) - \boldsymbol{x}$, where $\mathcal{H}(\boldsymbol{x})$ is desired underlying mapping. As shown in Fig. 2 (blue part and red part in the blue dash box), the weight-shared feature extractor G is constructed by residual connections, including nine convolutional layers and four basic residual blocks. Similarly, the projection layers P also consist of the shortcut connection. The detailed parameter settings of G and P are presented in Table I.

C. Joint MMD

As a kernel method, JMMD [27] has the ability to measure the joint distribution divergence. JMMD is the upper bound between the expectations of two joint distributions mapped by corresponding arbitrary functions ϕ defined in reproducing kernel Hilbert space $\mathcal{H}$. The divergence between $\mathcal{D}_s$ and $\mathcal{D}_t$ can be calculated by

$$\mathcal{L}_{\text{div}} = \left\| \mathbb{E}_{\mathcal{D}_s}\left(\otimes_{l=1}^{L} \phi^l\left(z_l^s\right)\right) - \mathbb{E}_{\mathcal{D}_t}\left(\otimes_{l=1}^{L} \phi^l\left(z_l^t\right)\right) \right\|^2_{\otimes_{l=1}^{L} \mathcal{H}^k} \quad (1)$$

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TABLE I
NETWORK PARAMETER SETTING

Layer	Outputs channel	Parameters or operation
Extractor		
Conv1	64	7×1kernel, stride2
Maxpool	-	3×1kernel, stride2
Conv2_x	64	[3×1kernel, stride2] ×2 Shortcut, Downsample
Conv3_x	128	[3×1kernel, stride2] ×2 Shortcut, Downsample
Conv4_x	256	[3×1kernel, stride2] ×2 Shortcut, Downsample
Conv5_x	512	[3×1kernel, stride2] ×2 Shortcut
Global Avepool	512	-
Classifier		
Fc1	No. of classes	Softmax
Projection Layers		
Conv1_x	512	[3×1kernel, stride2] ×3 Shortcut, Downsample

where L is the number of feature layers constituting the joint distribution, $\otimes$ is the mapping operation in product Hilbert space, and Z_l is the feature. In addition, multiple kernels [28] are used in JMMD instead of a single kernel to avoid the bias in estimation, and the convex combination of m kernels $\{k_\theta\}$ is written as

$$K \triangleq \left\{ k = \sum_{\theta=1}^{m} \lambda_\theta k_\theta : \sum_{\theta=1}^{m} \lambda_\theta = 1, \lambda_\theta \geq 0, \ \forall \theta \right\} \tag{2}$$

where λ_θ is the weight of the θth kernel. The proposed method uses JMMD for minimizing the joint distribution (features and predictions) divergence of the same-level fault features in the same-layer and cross-layer pairs.

III. PROPOSED METHOD

A. Adaptive Cross-Layer Alignment

The structure of the ACLAN-NC model is shown in the blue dashed box of Fig. 2. The network mainly includes feature extractor G, projection layers P, and classifier C. The first goal of ACLAN-NC is to initialize a discriminative model, and the cross-entropy classification loss based on source data can be denoted as

$$\mathcal{L}_{\text{cls}} = -\mathbb{E}_{(\mathbf{x}_i^s, y_i^s) \in \mathcal{D}_s} \sum_{c=1}^{|\mathcal{C}|} 1\{y_i = c\} \log p(y_i = c | \mathbf{x}_i) \tag{3}$$

where $p(\cdot)$ is the Softmax output, and y is the prediction.

The different-level features can be acquired from the corresponding m model layers of extractor G, denoted as $l^s = \{l_1^{s,i}, l_2^{s,i}, \ldots, l_m^{s,i}\}$ and $l^t = \{l_1^{t,j}, l_2^{t,j}, \ldots, l_m^{t,j}\}$ for ith source sample and jth target sample, respectively. Different from the multiple layer alignment methods, which directly align features of the same-level model layer and suffer from fault feature dislocation problem, ACLAN-NC aligns the same-level fault features for superior performance.

To realize cross-layer alignment, convolution-based projection layers P are introduced to map the features to the same size because the original features have different sizes. This process can be denoted as follows:

$$f_i^s = P^s\left(l_i^s\right), \ f_j^t = P^t\left(l_j^t\right) \tag{4}$$

where $P^s = \{P_1^s, P_2^s, \ldots P_m^s\}$ and $P^t = \{P_1^t, P_2^t, \ldots P_m^t\}$ are projection functions of different m layers for the source and target domains, respectively. The same size but different-level features $f_i^s = \{f_{i,1}^s, f_{i,2}^s, \ldots f_{i,m}^s\}$ and $f_j^t = \{f_{j,1}^t, f_{j,2}^t, \ldots f_{j,m}^t\}$ are obtained after projection layers.

To further mine the information hidden in each cross-layer pair, an adaptive allocation module is designed to assign weight to each feature pair, emphasizing the pair containing the same-level fault features and reducing feature dislocation. The features of source samples in batch B and target samples in another batch B will be averaged as $f_{\text{Avg}}^s = \sum_{i=1}^{B} f_i^s / B = \{f_1^s, f_2^s, \ldots f_m^s\}$ and $f_{\text{Avg}}^t = \sum_{j=1}^{B} f_j^t / B = \{f_1^t, f_2^t, \ldots f_m^t\}$, respectively. According to special similarity and channel similarity, the weight of each pair could be defined as

$$\omega_{p,n} = \frac{1}{2} \frac{\exp\left[\text{avg}\left(f_p^t \cdot \left(f_n^s\right)^{\text{T}}\right)\right]}{\sum_{\mu=1}^{m} \exp\left[\text{avg}\left(f_p^t \cdot \left(f_\mu^s\right)^{\text{T}}\right)\right]} + \frac{1}{2} \frac{\exp\left[\text{avg}\left(\left(f_p^t\right)^{\text{T}} \cdot f_n^s\right)\right]}{\sum_{\mu=1}^{m} \exp\left[\text{avg}\left(\left(f_p^t\right)^{\text{T}} \cdot f_\mu^s\right)\right]} \tag{5}$$

where the avg operator is to average the vectors. The 2-D (channel and hidden size) feature f has the size of $c \times h$. The first term of (5) reflects the spacial relationships by averaging the matrix with size $c \times c$, while the second term shows the channel relationships by averaging the matrix with size $h \times h$. The adaptive cross-layer alignment loss can, therefore, be denoted as

$$\mathcal{L}_{\text{cross}} = \sum_{p=1}^{m} \sum_{n=1}^{m} \omega_{p,n} \cdot \mathcal{L}_{\text{div}}\left(\left(f_p^t, y^t\right), \left(f_n^s, y^s\right)\right). \tag{6}$$

In addition, the distribution divergence minimization in the classifier (the final fully connected layer) could be regarded as the same-layer alignment

$$\mathcal{L}_{\text{same}} = \mathcal{L}_{\text{div}}\left(\left(l_c^t, y^t\right), \left(l_c^s, y^s\right)\right). \tag{7}$$

Finally, the unified alignment loss is minimized to constrain the same-level layer and different-level layer features

$$\mathcal{L}_{\text{ali}} = \delta \mathcal{L}_{\text{cross}} + (1 - \delta) \mathcal{L}_{\text{same}} \tag{8}$$

where the tradeoff hyperparameter δ is used to maintain a balance between cross-layer loss in the extractor and the same-layer loss in the classifier.

B. Batch Nuclear Norm Maximization and Minimization

Because of the distribution divergence, models trained under the supervision of source labels may lose discriminability in the target domain. The original prediction outputs are illustrated in the top black dash box of ACLAN-NC model in Fig. 2, from which it is clear that source samples and target samples are near the decision boundary, while some minority category samples (red pentagram) are misclassified.

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To avoid prediction uncertainty, Frobenius-norm (F-norm) maximization is used to guarantee the discriminability of the model, which can be denoted as

$$\|\mathrm{A}\|_F = \sqrt{\sum_{\tau=1}^{B}\sum_{\pi=1}^{|\mathcal{C}|}\left|\mathrm{A}_{\tau,\pi}\right|^2} \tag{9}$$

where A is the prediction matrix of classifier, B is the batchsize, and $|\mathcal{C}|$ is the total number of fault types. Based on the inequality of arithmetic and geometric means, the maximum of F-norm can be written as

$$\begin{aligned}\|\mathrm{A}\|_F &\leq \sqrt{\sum_{\tau=1}^{B}\left(\sum_{\pi=1}^{|\mathcal{C}|}\left|\mathrm{A}_{\tau,\pi}\right|\right)\cdot\left(\sum_{\pi=1}^{|\mathcal{C}|}\left|\mathrm{A}_{\tau,\pi}\right|\right)}\\ &= \sqrt{\sum_{\tau=1}^{B}1\cdot 1} = \sqrt{B}.\end{aligned} \tag{10}$$

In addition, some fault types are dominant in a random sampling batch, making the model with F-norm maximization tend to predict samples as the dominant categories. To avoid the collapse of the model into the majority categories, prediction matrix rank maximization is used to attain a more diverse model. The matrix rank is, however, difficult to optimize directly because it is an NP-hard nonconvex problem. According to [29], prediction diversity can be approximated as a nuclear norm, and we denote it as $\|\mathrm{A}\|_*$.

To strike a balance between discriminability and diversity, the potential association between F norm and nuclear norm is further explored as follows:

$$\frac{1}{\sqrt{D}}\|\mathrm{A}\|_* \leq \|\mathrm{A}\|_F \leq \|\mathrm{A}\|_* \leq \sqrt{D}\cdot\|\mathrm{A}\|_F \leq \sqrt{D\cdot B} \tag{11}$$

where $D = \min(B, |\mathcal{C}|)$. Equation (11) shows that $\|\mathrm{A}\|_F$ tends to be larger if $\|\mathrm{A}\|_*$ is increasing, meaning that a model with high discriminability and diversity can be attained by maximizing $\|\mathrm{A}\|_*$, and vice versa.

On the one hand, lessening the overly confident prediction on source domain batch B_S could be reached by reducing the distance from source samples to the decision boundary, i.e., minimizing the nuclear norm of source data. On the other hand, maximizing the nuclear norm on the target data batch B_T improves discriminability and diversity simultaneously. The final restriction of prediction can, therefore, be formulated as

$$\mathcal{L}_B = \frac{1}{B_S}\|\mathrm{A}_S\|_* - \frac{1}{B_T}\|\mathrm{A}_T\|_*. \tag{12}$$

As shown in the bottom black dash box of the ACLAN-NC model of Fig. 2, after nuclear-norm constraints, the prediction boundary is more reliable because the samples are away from it. In addition, the prediction tendency to majority categories can be corrected, and the minority samples are correctly classified. In summary, good transfer priorities, i.e., discriminability and diversity, are guaranteed in the domain adaptation process.

C. Overall Objective and Optimization

The overall optimization goals include classification loss $\mathcal{L}_{\mathrm{cls}}$ based on source domain labels, alignment loss $\mathcal{L}_{\mathrm{ali}}$, and nuclear-norm constraints $\mathcal{L}_B$ on predictions. Finally, the training optimization objective of ACLAN-NC is as follows:

$$\mathcal{L} = \mathcal{L}_{\mathrm{cls}} + \Lambda\mathcal{L}_{\mathrm{ali}} + \beta\mathcal{L}_B \tag{13}$$

where the tradeoff parameter Λ and β help balance each part of $\mathcal{L}$, Λ is chosen according to the formula [30]

$$\Lambda = \alpha \times \left(\frac{2}{1+\exp(-10\times q)} - 1\right) \tag{14}$$

where q starts at 0 and increases linearly to 1 during the training process.

Algorithm 1 Training Procedure for ACLAN-NC

Input: Source samples $(\mathbf{x}_i^s, y_i^s)$ and target samples $\mathbf{x}_j^t$; Tradeoff parameters: α, β, and δ; Batch-size: B; Max epoch: T.
Output: Well-trained extractor G and classifier C.
Initialize G, C, and P.
For epoch ← 1 to T do
Random sample B source samples and B target samples in a mini-batch.
Step1: Extract different m level source fault features $l_i^s = \{l_1^{s,i}, l_2^{s,i}, \ldots, l_m^{s,i}\}$ and target fault features $l_j^t = \{l_1^{t,j}, l_2^{t,j}, \ldots, l_m^{t,j}\}$ from G.
Step2: Calculate the cross-entropy loss on source samples according to (3).
Step3: Using projection function P to obtain the same size but different-level features $f_i^s = \{f_{i,1}^s, f_{i,2}^s, \ldots f_{i,m}^s\}$ and $f_j^t = \{f_{j,1}^t, f_{j,2}^t, \ldots f_{j,m}^t\}$ according to (4).
Step4: Calculate the weight of each feature pair according to (5) and obtain the cross-layer alignment loss according to (6).
Step5: Use features l_c^s and l_c^t in classifier C to obtain same layer alignment loss according to (7).
Step6: Restrict the prediction to keep diversity and discriminability according to (12).
Step7: Optimize G, C, and projections layers P by minimizing all losses according to (13).
End for

IV. Experiments

A. Datasets

Case 1 (PHM 2009 Dataset): The industrial gearbox is important in manufacturing equipment, and the public industrial gearbox dataset PHM2009 [31], [32] is used to conduct the first set of experiments. The experiment platform mainly includes four gears, three axes, and six bearings, as presented in Fig. 3. The helical gears with six fault types (denoted as H1, H2, H3, H4, H5, and H6) in PHM2009 are used to construct the varying speed experiment. The accelerator is attached to input shaft retaining plates, and the sampling frequency is 66.7 kHz.

PHM2009 has five different rotating speeds, including 1800, 2100, 2400, 2700, and 3000 r/min. For a proper number of experiments, the first four speeds with equal variation, i.e., 1800, 2100, 2400, and 2700 r/min, are used as four different domains (L_0, L_1, L_2, and L_3) to construct twelve diagnosis

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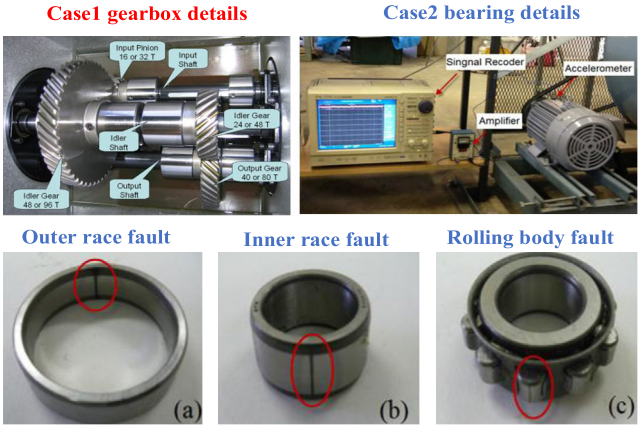

Fig. 3. Inside the gearbox of Case 1 and the bearing experiment platform details of Case 2. (a) Outer race fault. (b) Inner race fault. (c) Rolling body fault.

tasks. Transfer fault diagnosis tasks under the first four rotating speeds are more challenging than the last four because the relative change in speed of the first four is (2700–1800)/1800, i.e., 50%, which is larger than that of the last four (43%). Each type of fault contains 400 samples, of which 200 samples are used as training sets, 100 samples are applied as test sets, and 100 samples are used as validation sets. There are 2048 data points in each sample.

Case 2 (JiangNan University (JNU) Dataset): The second experiment dataset is from JiangNan University [33]. The bearing experiment under varying working conditions was built under three different rotating speeds, and the sampling frequency was 50 kHz. As shown in Fig. 3, there are four different types of test bearings, including normal (NA), inner fault (IF), out fault (OF), and rolling body fault (BF). As illustrated in Table II, N205 rolling bearings are applied to implement normal, outer fault, and rolling body fault, while NU205 rolling bearings are used for inner fault. Three rotating speeds of 600, 800, and 1000 r/min construct three different domains (B_0, B_1, and B_2). Each sample contains 1024 points, and there are 400 samples for each fault type under each working condition. Finally, six transfer learning tasks are constructed for performance evaluation.

Case 3 (Practical Bearing Fault Test Rig PT500): The third dataset is collected from a real fault diagnosis platform PT500. The platform and different fault bearings are shown in Fig. 4. Five working conditions, which are denoted as A_0, A_1, A_2, A_3, and A_4, are constructed according to five rotating speeds of 800, 1200, 1600, 2000, and 2400 r/min. The fault categories include normal, ball fault, outer race fault, inner race fault, and compound fault combined with both outer and inner race faults. The four tasks are established by A_0 as the source domain, while A_1, A_2, A_3, and A_4 are target domains, respectively.

B. Baseline Methods

1) ResNet: The commonly used backbone of the classification network ResNet is built as a basic comparing approach. The source samples are used to initialize a discriminative model, and then target samples are directly fed into the model for testing. ResNet has the same structure as ACLAN-NC, except that there is no domain adaptation module. All comparison methods use ResNet as the backbone.

TABLE II
SIZE OF FAULTS IN JNU DATASET

Contents	N205	NU205
Bearing outer diameter	52 mm	52 mm
Bearing inner diameter	25 mm	25 mm
Bearing width	15 mm	15 mm
Bearing roller diameter	7 mm	7 mm
Number of the rollers	10	11
Outer race defect (width × depth)	0.3×0.25 mm	-
Rolling element defect (width × depth)	0.3×0.15 mm	-
Inner race defect (width × depth)	-	0.3×0.25 mm

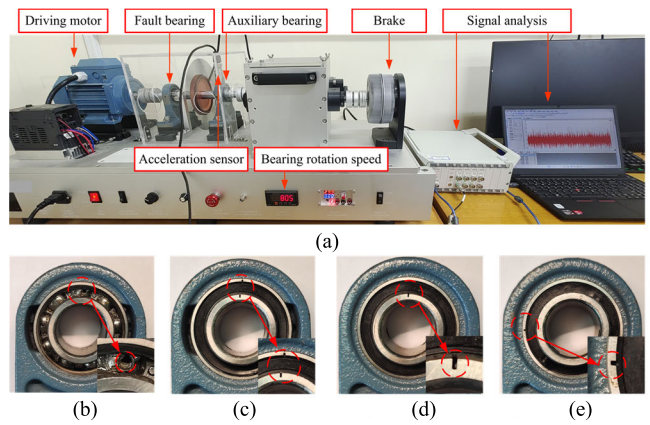

Fig. 4. Real bearing fault diagnosis testbed PT500 and fault bearings. (a) Real bearing experiment platform. (b) Ball fault. (c) Compound fault. (d) Inner race fault. (e) Outer race fault.

2) CORAL: As another widely applied metric-based adaptation method, CORAL [12] measures the domain divergence by calculating second-order statistics information.

3) DANN: In DANN, a domain discriminator is constructed to distinguish whether the output of the feature extractor is from the source domain or target domain, and thus, the extractor is forced to extract domain-invariant features [14].

4) CDANE: Conditional adversarial domain adaptation with entropy (CDANE) is used for minimizing domain divergence. Feature information and label information after multilinear map operation are both used in CDANE [34], and the same category features from different domains could be aligned.

5) Mixed Methods: To make full use of the superiority of metric-based alignment and adversarial-based alignment methods, a mixed method is designed. CORAL and CDANE are used to construct the joint adaptation objective function.

6) AACD: Adversarial adaptation-based classifier discrepancy [35] uses two classifiers to identify the misaligned region. The classifiers are updated to maximize the prediction divergence on the target domain, and then the extractor is updated to minimize the divergence. By the adversarial process between the classifiers and extractor, AACD can learn domain-invariant features.

7) DALN: The fault classifier in DALN is reused as the domain discriminator to achieve feature alignment [36]. The nuclear-norm Wasserstein discrepancy, which measures

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TABLE III
CLASSIFICATION ACCURACY OF PHM2009 DATASET (%)

Task	ResNet	CORAL	DANN	CDANE	Mixed	AACD	DALN	TPC	ACLAN-NC
$L_0 \rightarrow L_1$	64.25±2.16	65.42±3.71	72.51±4.09	78.06±4.03	76.55±4.00	53.78±19.5	**83.60±10.5**	72.31±14.3	79.03±4.54
$L_0 \rightarrow L_2$	56.68±2.93	60.68±5.55	70.91±5.73	70.10±7.71	83.70±2.19	38.30±10.6	85.66±2.52	79.61±7.19	**85.77±3.59**
$L_0 \rightarrow L_3$	38.97±1.83	51.46±5.48	44.59±9.95	41.37±1.46	41.13±11.6	36.72±9.83	**75.44±11.1**	69.05±17.8	65.57±7.07
$L_1 \rightarrow L_0$	62.43±2.04	67.52±2.07	67.38±1.29	69.88±3.79	68.32±1.12	69.65±2.29	70.83±4.21	**72.77±6.80**	69.29±4.81
$L_1 \rightarrow L_2$	70.36±1.07	75.85±2.26	76.13±2.44	80.70±2.11	**82.18±1.21**	78.65±0.96	80.40±1.47	80.33±2.43	80.70±1.04
$L_1 \rightarrow L_3$	55.03±2.30	62.68±2.31	61.73±1.92	69.46±4.66	65.93±2.84	66.26±12.0	67.57±1.51	**69.77±2.74**	65.87±1.92
$L_2 \rightarrow L_0$	55.48±1.83	76.19±3.10	86.88±2.23	91.51±7.51	94.14±1.01	39.72±9.23	93.21±1.55	94.92±2.70	**97.48±0.98**
$L_2 \rightarrow L_1$	70.58±1.53	76.21±1.46	75.89±1.41	79.46±1.32	80.22±1.66	78.00±2.70	81.00±1.82	80.25±2.96	**81.28±1.28**
$L_2 \rightarrow L_3$	71.53±2.01	79.48±3.83	89.28±2.17	94.93±1.48	95.67±1.16	91.55±6.95	95.51±2.35	82.06±14.2	**97.18±0.47**
$L_3 \rightarrow L_0$	25.92±2.14	49.90±3.50	70.51±5.58	53.13±11.6	60.36±25.4	16.90±0.19	74.77±4.38	67.24±8.37	**88.18±4.59**
$L_3 \rightarrow L_1$	44.24±3.26	60.28±2.88	72.84±1.45	69.91±2.80	74.23±1.12	26.77±8.25	74.32±3.31	61.15±7.63	**76.48±1.38**
$L_3 \rightarrow L_2$	58.78±1.22	78.30±1.22	79.93±2.81	79.17±3.83	87.00±1.52	49.62±14.5	86.73±3.30	77.73±9.53	**88.16±1.40**
Average	56.19±2.03	67.00±3.11	72.38±3.42	73.14±4.36	75.79±2.66	53.83±8.09	80.76±3.21	75.60±8.06	**81.25±2.76**

the discriminability, is ensembled in classifiers to perform as the discriminator and avoid the requirement of gradient penalty or weight clipping.

8) TPC: Toward prediction constraints [18] minimizes the class confusion of the target domain and restricts the prediction nuclear norm to balance discriminability and diversity for good fault diagnosis performance on transfer tasks.

C. Implementation Details

All the algorithms are implemented based on PyTorch, and each model is trained for 200 epochs. Minibatch Adam optimizer is adopted, and there are 64 samples in a batch. The learning rate is 0.001 at the beginning and then multiplies by 0.1 in epochs 100 and 150, respectively. For each method, randomly repeat the experiments ten times to obtain average accuracy and standard deviation.

D. Results on PHM2009 Dataset

1) Classification Results Analysis: Table III and Figs. 5 and 6 report the detailed classification results of comparison methods. It is obvious that ACLAN-NC outperforms the comparison methods and attains the highest accuracy of 81.25%. Among the eight comparison methods, DALN obtained the best result of 80.76%, but its performance on nine tasks is worse than ACLAN-NC. It is also worth mentioning that among the 12 transfer tasks, ACLAN-NC reaches the highest diagnosis accuracy in seven tasks. In particular, for task $L_3 \rightarrow L_0$, basic ResNet only has 25.92% classification accuracy, and about half of the samples are misclassified by CDANE. In contrast, the proposed method significantly obtains better results. As illustrated in Fig. 6, the diagnosis accuracy of AACD on transfer fault diagnosis tasks $L_2 \rightarrow L_1$ and $L_2 \rightarrow L_3$ is competitive, but AACD has unsatisfactory performance on $L_0 \rightarrow L_2$, $L_0 \rightarrow L_3$, $L_2 \rightarrow L_0$, $L_3 \rightarrow L_0$, and $L_3 \rightarrow L_1$, which pulls down the average accuracy of AACD. In addition, ACLAN-NC is more stable than most comparison transfer learning methods, which can be reflected by the low standard deviation of 2.76%.

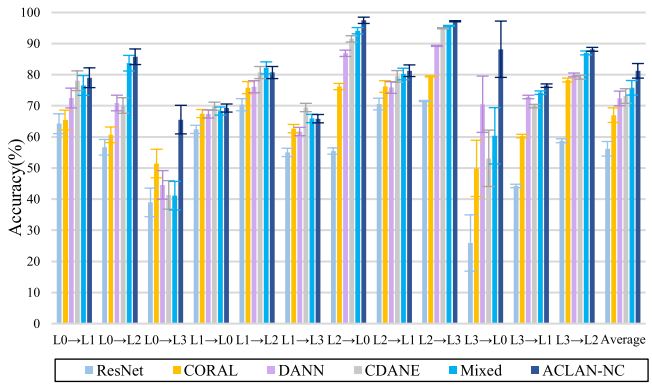

Fig. 5. Classification results of ResNet, CORAL, DANN, CDANE, Mixed, and ACLAN-NC on PHM2009 dataset.

2) Feature Visualization: Feature visualization results using t-SNE further demonstrate the superiority of ACLAN-NC. The feature distribution results, which are randomly selected in ten tests on the task $L_3 \rightarrow L_0$, are visualized in Fig. 7. It is observed that domain-invariant fault features are learned by ACLAN-NC; especially, features of the same category from source and target domains are well aligned, and different types of faults are well separated. In contrast, according to Fig. 7, the comparison methods cannot accurately align features of the same category, causing severe performance degradation.

For example, even the mixed method still incorrectly identifies H6 as H1 and H2, while ACLAN-NC successfully classifies most H6 samples. In summary, the results validate that ACLAN-NC is able to address the fault diagnosis tasks.

3) Confusion Matrix Analysis: The confusion matrices on task $L_2 \rightarrow L_0$ are used to analyze the classification performance of different methods. Fig. 8(a) shows that the regular ResNet exhibits serious misclassification, making it difficult to distinguish the samples of categories H1, H2, and H6. A similar phenomenon occurs in CORAL and DANN, as shown in Fig. 8(b) and (c), respectively. CDANE and mixed method also

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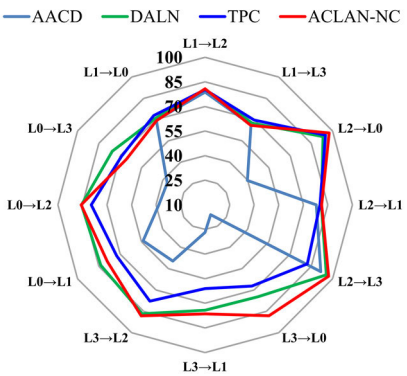

Fig. 6. Classification results of AACD, DALN, TPC, and ACLAN-NC on PHM2009 dataset.

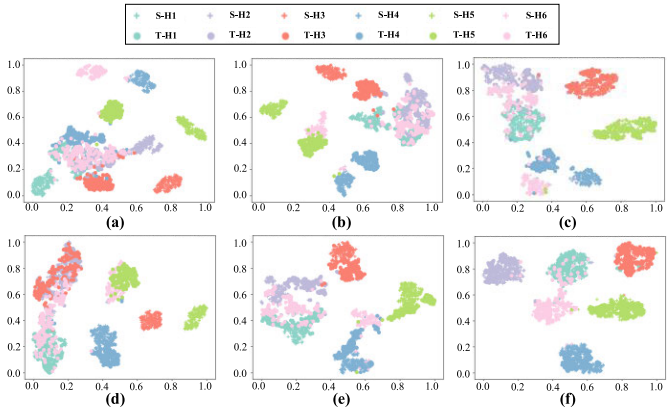

Fig. 7. Feature visualization results of PHM2009 dataset. (a) ResNet. (b) CORAL. (c) DANN. (d) CDANE. (e) Mixed. (f) ACLAN-NC.

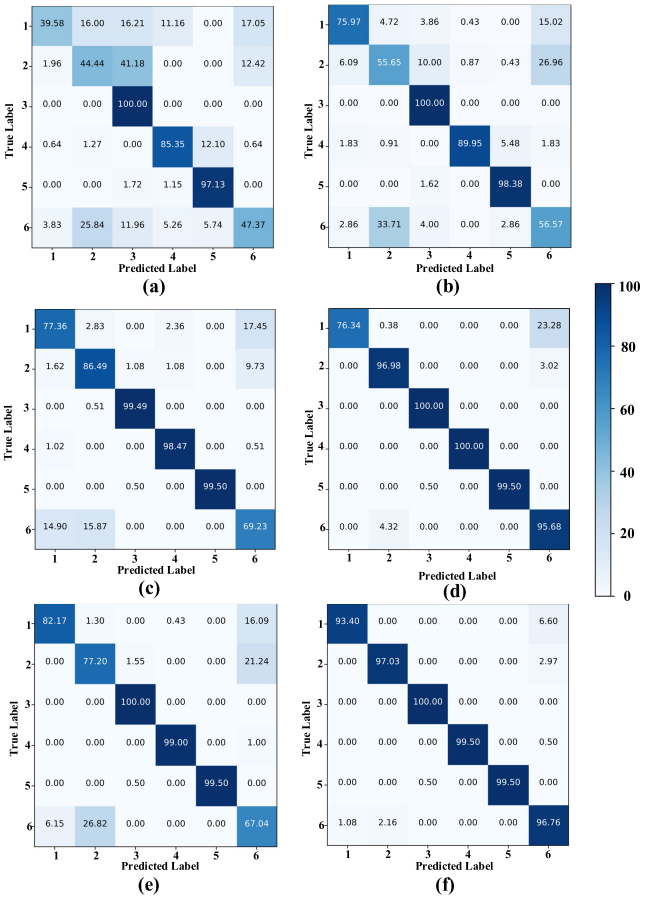

Fig. 8. Confusion matrix analysis (%). (a) ResNet. (b) CORAL. (c) DANN. (d) CDANE. (e) Mixed. (f) ACLAN-NC.

have different degrees of classification confusion. In contrast, Fig. 8(f) shows that ACLAN-NC can significantly improve recognition accuracy.

4) Sensitivity Analysis: Based on the experimental results of randomly selected tasks in PHM2009, Fig. 9 reports the sensitivity of hyperparameters δ, α, and β. First, δ is varied within $\{0.1, 0.3, 0.5, 0.7, 0.9\}$. It is found that ACLAN-NC achieves the best diagnosis results when δ is equal to 0.7, and the model is generally robust to δ. Similarly, α is set to vary within $\{0.1, 0.3, 0.5, 0.75, 1\}$. The results on different tasks of different values of α are shown in Fig. 9(b). There is a peak at $\alpha = 0.3$, and the classification accuracy remains generally stable in the range of α from 0.1 to 0.75. Finally, β is varied within $\{0.1, 0.25, 0.5, 0.75, 1\}$ to show its influence. The classification accuracy remains generally stable in the range of β from 0.1 to 0.5, and the best performance is attained at $\beta = 0.25$. The above analysis indicates ACLAN-NC is generally robust, and exhaustive hyperparameter fine-tuning can be avoided.

E. Results on the JNU Dataset

1) Classification Results Analysis: Table IV summarizes the classification result on the JNU bearing dataset, and several observations could be concluded. In contrast to the comparison methods, ACLAN-NC achieves the highest classification accuracy rate of 95.95%. In addition, the proposed method obtains three best results on six transfer tasks, while TPC and DANN achieve two and one optimal results, respectively. It is obvious that CORAL performs worse than basic ResNet because fixed metrics may cause negative transfer. Among the comparison methods, CDANE is capable of getting competitive results, but ACLAN-NC still has a higher accuracy and smaller standard deviation. In task $B_1 \rightarrow B_0$, basic ResNet only has 65.91% classification accuracy, while CORAL and the mixed method increase this by 8% and 18%, respectively; moreover, DANN, CDANE, and DALN achieve competitive classification accuracy of more than 90% on the same task, but their performance is still worse than ACLAN-NC. The above observations confirm the superiority and robustness of ACLAN-NC.

2) Feature Visualization: One result of transfer task $B_1 \rightarrow B_0$ is randomly selected to carry out the feature visualization experiment in Fig. 10. For ResNet, CORAL, and mixed methods, it is evident that the features from different domains are not well aligned; the negative transfer caused by CORAL can, especially, be observed in Fig. 10(b) and (e). The features from different domains belonging to the same fault type are not properly clustered, and the features of diverse fault types are mixed together, resulting in a low diagnosis accuracy. At the same time, according to Fig. 10(d) and (f), CDANE and ACLAN-NC successfully align the domain divergence, indicating those methods learn more domain-invariant features; however, there are still some overlapping areas in CDANE;

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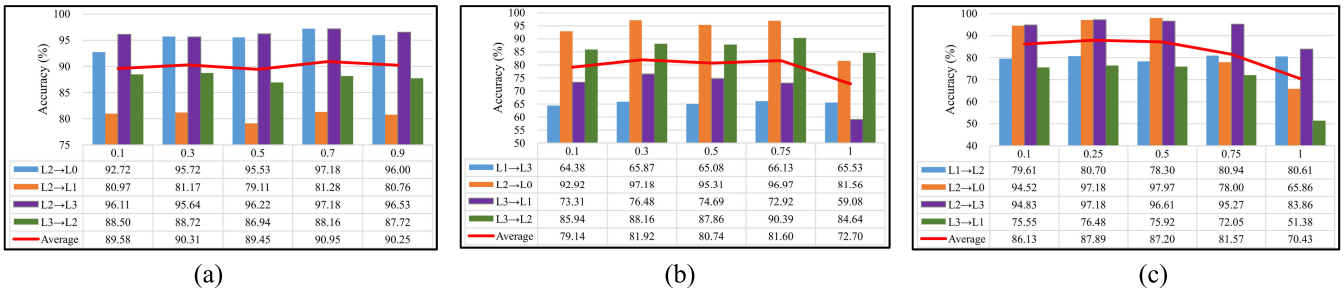

Fig. 9. Sensitivity analysis of the hyperparameters on the PHM2009 dataset. (a) Sensitivity analysis of δ. (b) Sensitivity analysis of α. (c) Sensitivity analysis of β.

TABLE IV
CLASSIFICATION ACCURACY OF JNU DATASET (%)

Task	ResNet	CORAL	DANN	CDANE	Mixed	AACD	DALN	TPC	ACLAN-NC
$B_0 \rightarrow B_1$	92.50±1.62	88.28±3.84	98.28±0.81	98.31±0.86	92.41±2.81	98.97±0.38	98.66±1.07	99.03±0.20	**99.13±0.31**
$B_0 \rightarrow B_2$	83.62±1.95	87.31±3.45	94.59±0.86	95.63±0.57	89.91±1.68	97.13±0.65	94.53±1.79	**97.38**±0.64	96.62±0.70
$B_1 \rightarrow B_0$	65.91±1.86	73.56±5.36	92.84±1.21	91.44±1.62	83.59±4.57	88.31±10.4	90.56±2.00	89.97±1.66	**93.12±1.59**
$B_1 \rightarrow B_2$	95.37±1.13	94.47±1.22	97.16±0.55	98.28±0.53	94.53±2.15	98.25±0.38	97.88±0.48	**98.34±0.37**	97.94±0.50
$B_2 \rightarrow B_0$	82.00±2.54	67.99±6.99	**90.88±1.12**	90.53±1.40	80.03±4.19	90.31±1.09	88.66±2.17	88.63±1.27	89.34±1.71
$B_2 \rightarrow B_1$	97.22±1.00	91.44±1.91	98.84±0.52	99.16±0.40	92.22±3.00	99.31±0.27	93.34±16.3	99.33±0.09	**99.53±0.21**
Average	86.10±1.68	83.84±3.80	95.43±0.85	95.55±0.90	88.78±3.07	95.38±2.19	93.94±3.96	95.45±0.71	**95.95±0.83**

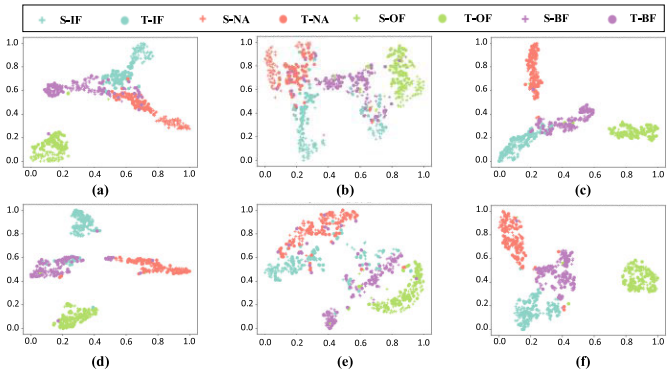

Fig. 10. Feature visualization results of JNU dataset. (a) ResNet. (b) CORAL. (c) DANN. (d) CDANE. (e) Mixed. (f) ACLAN-NC.

TABLE V
CLASSIFICATION ACCURACY OF DIFFERENT PROJECTION STRUCTURES (%)

Layers	Residual block	Average
1	√	98.78
1	×	98.44
2	√	98.87
2	×	98.22
3	√	**99.13**
3	×	98.13

TABLE VI
CLASSIFICATION ACCURACY WITH DIFFERENT NUMBERS OF ALIGNMENT LAYERS (%)

Number of layers for alignment	2	3	4
$B_2 \rightarrow B_0$	89.06	86.56	**89.34**
$B_2 \rightarrow B_1$	99.06	99.38	**99.53**
$L_2 \rightarrow L_0$	95.75	91.58	**97.48**
$L_2 \rightarrow L_1$	73.66	81.00	**81.28**
Average	89.38	89.63	**91.91**

inner faults and rolling body faults are, especially, misclassified. In summary, the results manifest the better transfer ability of the proposed method.

3) Structure of Projection Network: The projection networks in ACLAN-NC consist of three residual convolution layers. Different structures of the projection layer are compared under task $B_0 \rightarrow B_1$ on the JNU dataset in Table V. It is clear that the residual connection in projection layers could improve classification accuracy. Overall, different convolution layers and residual block settings have little effect on the diagnosis accuracy, and the three-layer residual convolution projection networks are adopted in ACLAN-NC to achieve better results.

4) Number of Alignment Layers: In order to study the influence of different layers for alignment, Table VI compares the classification accuracy based on four tasks. The results reveal that as additional layers provide extra information for alignment, the classification accuracy increases when the number of alignment layers rises; therefore, the number of layers for alignment is set to four in ACLAN-NC.

5) Sensitivity Analysis: Fig. 11 displays the sensitivity of three parameters on the JNU dataset. The model has the highest accuracy when δ is 0.3, and there is a decreasing trend when δ is larger than 0.3. As indicated in Fig. 11(b), α is set to 0.3 for best diagnosis performance. The highest accuracy arrives when β is 0.5, and the proposed method is generally robust when β is less than 0.75. As shown in Fig. 11(c), there

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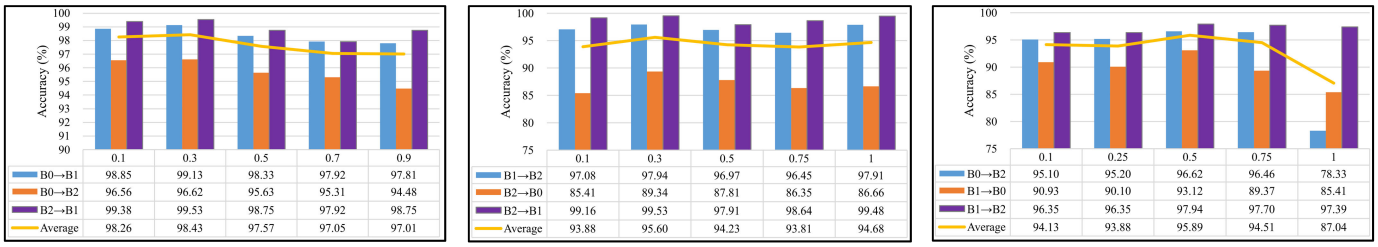
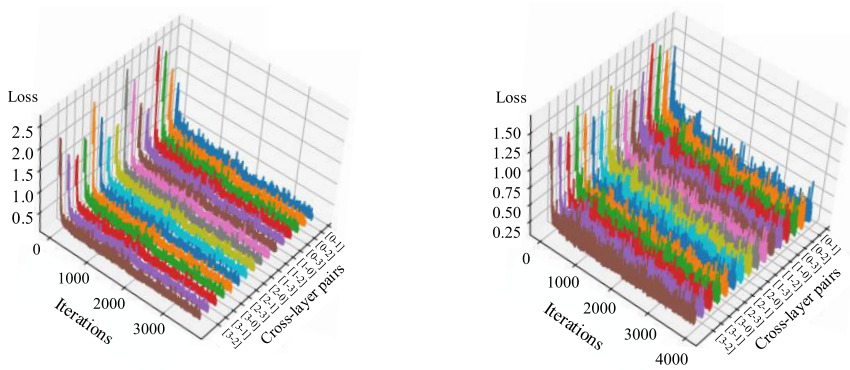


(a) (b) (c)

Fig. 11. Sensitivity analysis of the hyperparameters on the JNU dataset. (a) Sensitivity analysis of δ. (b) Sensitivity analysis of α. (c) Sensitivity analysis of β.



(a) (b)

Fig. 12. Convergence curves of multiple JMMD losses on task $L_0 \rightarrow L_3$ and task $B_2 \rightarrow B_1$. (a) Convergence curves on $L_0 \rightarrow L_3$. (b) Convergence curves on $B_2 \rightarrow B_1$.

is a significant reduction trend when β is larger than 0.75, which is similar to the phenomenon in the PHM2009 dataset.

6) Convergence of Different Losses: The curves of multiple JMMD losses on randomly selected task $L_0 \rightarrow L_3$ and task $B_2 \rightarrow B_1$ are displayed in Fig. 12(a) and (b), respectively, where the X-axis means the iteration number of batches, Y-axis means different cross-layer pairs, and Z-axis means the alignment loss. There are 16 cross-layer pairs because the number of layers for alignment is set to four. It is clear that all the JMMD losses converge during the training process. In fact, when a JMMD loss is reduced, the feature distribution divergence decreases, which means the feature extractor could obtain more domain-invariant features. The convergence curves in Fig. 12 illustrate that the training loss curves of different cross-layer pairs show a synchronous downward trend, indicating that the multiple JMMD losses converge when trained together. The results further indicate the stability and robustness of ACLAN-NC.

7) Alignment Effect of Original Features: To measure the alignment effect of original features, the difference between the alignment loss (JMMD) of features after projections minus the alignment loss of the original features is used as the evaluation metric of the original feature alignment. Based on task $B_2 \rightarrow B_0$, Fig. 13 displays the alignment effect of original features in the four different same-level layers, respectively. The difference between losses will not strictly converge to zero because we reduce the distribution divergence by minimizing the alignment loss of projection layer features and achieving same-level feature alignment. The alignment of original features is not directly implemented to avoid feature dislocation, which corroborates with the result that

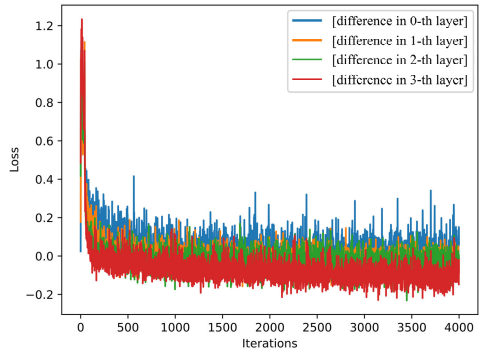


Fig. 13. Alignment process of original features.

the alignment loss of features after projection is not strictly equal to the alignment loss of original features; however, it is noticeable that the alignment of original features and the alignment of features after projections are generally synchronized because the difference is close to zero after 500 iterations. It is clear that the original features are aligned as the alignment is achieved in the projection layers.

F. Results on Practical PT500 Dataset

1) Classification Results Analysis: The classification results of comparison methods are shown in Table VII. The proposed method achieves the highest average accuracy and outperforms the comparison methods on tasks $A_0 \rightarrow A_3$ and $A_0 \rightarrow A_4$. Compared to the competitive mixed method, DALN, and TPC, ACLAN-NC has a lower standard deviation, proving the robustness of the proposed method. Overall, the transfer

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TABLE VII
CLASSIFICATION ACCURACY OF PT500 DATASET (%)

Task	ResNet	CORAL	DANN	CDANE	Mixed	AACD	DALN	TPC	ACLAN-NC
$A_0 \rightarrow A_1$	82.56±2.64	95.30±1.06	81.30±19.4	**96.04±5.30**	89.50±8.11	75.80±6.40	93.30±2.40	95.48±5.17	95.04±1.72
$A_0 \rightarrow A_2$	74.50±3.84	88.26±3.55	83.54±19.6	89.18±10.7	92.80±5.37	69.24±10.9	**96.44±2.15**	95.22±8.45	95.38±5.72
$A_0 \rightarrow A_3$	65.56±3.11	78.72±2.22	77.76±18.6	83.16±13.6	90.38±9.71	61.40±11.8	83.96±11.8	81.94±11.3	**94.46±2.17**
$A_0 \rightarrow A_4$	50.76±4.14	71.34±4.59	81.20±8.41	59.96±23.6	88.28±7.36	49.56±13.7	68.06±6.52	72.64±11.8	**88.48±7.11**
Average	68.35±3.43	83.41±2.86	80.95±16.5	82.09±10.3	90.24±7.64	64.00±10.7	85.44±5.72	86.32±9.18	**93.34±4.18**

TABLE VIII
ABLATION STUDY OF THE PROPOSED METHOD (%)

Dataset	w/o $\mathcal{L}_B$	w/o $\mathcal{L}_{\mathrm{Bs}}$	w/o $\mathcal{L}_{\mathrm{Bt}}$	w/o $\mathcal{L}_{ali}$	Ours
PHM2009	73.98	80.27	79.30	79.70	**81.25**
PT500	88.91	85.91	82.12	88.79	**93.34**

diagnosis tasks become more difficult as the rotation speed difference between the source domain and target domain increases. For example, the accuracy of TPC on task $A_0 \rightarrow A_1$ and task $A_0 \rightarrow A_2$ exceeds 95%, but the performance degrades as the greater domain divergence. The diagnosis accuracy on task $A_0 \rightarrow A_3$ and task $A_0 \rightarrow A_4$ of ACLAN-NC are 94.46% and 88.48%, respectively, which are higher than all comparison methods. It can be summarized that ACLAN-NC is capable of solving fault diagnosis tasks under different working conditions.

2) Ablation Studies: Ablation studies are accomplished on PHM2009 and PT500 datasets to further investigate the effectiveness of each part of ACLAN-NC. Table VIII shows the detailed results. First, according to the final optimization goal of ACLAN-NC, batch nuclear-norm constraint $\mathcal{L}_B$ is removed to construct ACLAN-NC without $\mathcal{L}_B$ (w/o $\mathcal{L}_B$). Compared with ACLAN-NC, the classification accuracy without $\mathcal{L}_B$ drops to 73.98% and 88.91% for PHM2009 and PT500, respectively, showing the importance of batch nuclear-norm constraints; especially, prediction constraints on the source domain batch B_S and target domain batch B_T are removed to implement ACLAN-NC without $\mathcal{L}_{\mathrm{Bs}}$ (w/o $\mathcal{L}_{\mathrm{Bs}}$) and ACLAN-NC without $\mathcal{L}_{\mathrm{Bt}}$ (w/o $\mathcal{L}_{\mathrm{Bt}}$), respectively. The classification results show that both source and target domain constraints are required. In addition, cross-layer alignment loss $\mathcal{L}_{\mathrm{ali}}$ is removed to construct ACLAN-NC without $\mathcal{L}_{\mathrm{ali}}$ (w/o $\mathcal{L}_{\mathrm{ali}}$). The classification accuracy is reduced to 79.70% and 88.79% for the two datasets, respectively. Hence, the cross-layer relationship between different domains is necessary for domain adaptation. These phenomena indicate that whichever loss item is removed, the model will have performance degradation.

3) Computation Cost and Online Testing Efficiency: To analyze the efficiency of the proposed method, the training time and testing time of each batch on the PT500 dataset are displayed in Table IX. The results are tested on a computer with Core i9 CPU, GeForce RTX4090 GPU, and 64GB RAM. As shown in Table IX, the test time of different methods is almost the same. Since the proposed method aligns features of different levels, ACLAN-NC has a longer training time than some comparison methods but improves the fault diagnosis performance by using multilevel features. Overall, the high accuracy and efficient testing response of ACLAN-NC guarantee the online monitoring of bearing working conditions.

TABLE IX
TRAINING TIME AND TEST TIME OF EACH BATCH ON PT500 DATASET USING DIFFERENT METHODS

Different methods	Training time (10^{-1} s)	Test time (10^{-2} s)
ResNet	2.94	1.06
CORAL	2.95	1.08
DANN	2.97	1.08
CDANE	2.99	1.06
Mixed	3.00	1.09
AACD	9.23	1.09
DALN	2.99	1.07
TPC	2.96	1.10
ACLAN-NC	4.85	1.08

TABLE X
COMPARISON RESULTS OF MULTILAYER ALIGNMENT METHODS (%)

Dataset	PHM2009	JNU	PT500
Λ_1	76.98	94.98	**96.13**
Λ_2	68.05	90.28	86.67
Λ_3	72.72	91.72	92.00
Λ_4	50.78	92.05	89.05
ACLAN-NC	**81.25**	**95.95**	93.34

4) Comparison With Multilayer Alignment Methods: The transfer learning methods with multilayer alignment loss generally have better performance than the methods that only align the single layer [37]. In order to further validate the performance of the proposed method, different multilayer alignment methods are built for comparison. The fault features that had been used in ACLAN-NC are used to construct the multilayer alignment methods, as detailed below. It should be noted that MK-MMD loss is used as the metric, i.e., $\mathcal{L}_{\mathrm{DAN}}$. The prediction constraint loss, which is used to balance the discriminability and diversity of prediction in our method, is also added to the final loss function of each comparison method for a fair comparison. The first multilayer alignment comparison method is built by aligning l_c and l_4 simultaneously, and the other methods gradually align more layers on the previous methods. The losses of different multilayer alignment methods are expressed as

$$\mathcal{L}_M = \sum_{m=1}^{M} \mathcal{L}_{\mathrm{DAN}}\left(l_{5-m}^t, l_{5-m}^s\right) + \mathcal{L}_{\mathrm{DAN}}\left(l_c^t, l_c^s\right) + \mathcal{L}_{\mathrm{cls}} + \beta \mathcal{L}_B \tag{15}$$

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where M is varied within $\{1, 2, 3, 4\}$. The classification results are shown in Table X.

As Table X shows, ACLAN-NC achieves the best performance on PHM2009 and JNU datasets, and there is no method to achieve the best performance consistently. It can be seen that the number of aligned layers does not necessarily result in performance improvement. Since the optimal combination of multilayer alignment is not available in advance, the hand-craft multilayer alignment may cause performance degradation. ACLAN-NC outperforms the comparison methods due to adaptively aligning feature pairs, but the inferior results on PT500 indicate that there is room for a cross-layer alignment strategy.

V. Conclusion

The feature dislocation problem could lead to the degradation of fault diagnosis performance because aligning features only within the same layer may cause negative transfer. Generally, the same-level fault features are possibly distributed across the different layers of the model; however, feature relationships between different layers are ignored by the majority of existing fault diagnosis methods. In response to this new challenge, adaptive cross-layer alignment is introduced into the proposed method; furthermore, the nuclear norm of prediction is minimized and maximized for source and target domains, respectively, to guarantee strong transferability of fault features. A novel IFD approach, ACLAN-NC, is constructed to improve the classification accuracy from different perspectives. Finally, through extensive experiments, the superiority of ACLAN-NC is fully verified.

In the future, the cross-layer alignment will be extended into the more common scenario, where we have multiple source domains and multiple target domains. It can be expected that using cross-layers to align features in each source domain and each target domain will provide more knowledge to extract the domain invariant features. In addition, taking vision transformer architecture as the extractor, cross-layer alignment can be achieved by using more shallow projection layers or even without projection functions, which further improves the accuracy of fault diagnosis and reduces the training time.

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